

Kernel Discriminant Analysis and Nearest Neighbor Methods

Statistical Learning -I

Lectures: September 28 & September 30, 2026

1 Kernel Discriminant Analysis (KDA)

1.1 Definition

$$\delta_{KDA}(x) = \operatorname{argmax}_{1 \leq i \leq J} \pi_i \hat{f}_{i,h_i}(x)$$

or $\operatorname{argmax}_{1 \leq i \leq J} \pi_i \hat{f}_{ih}(x)$, if a common bandwidth h is used for all classes.

We have seen that if $h \rightarrow 0$ and $nh^d \rightarrow \infty$ as $n \rightarrow \infty$, then for any fixed x ,

$$\hat{f}_h(x) \xrightarrow{P} f(x).$$

One can show that for such a choice of h , the error rate of the KDA rule converges to the Bayes risk.

But what about its behavior for fixed sample size as $h \downarrow 0$? (Assume that the Gaussian kernel is used for density estimation.)

$$X_{11}, X_{12}, \dots, X_{1n_1} \quad X_{21}, X_{22}, \dots, X_{2n_2}$$

Here x denotes the query point, and we consider KDA with a very small h .

$$\pi_1 \hat{f}_{1h}(x) > \pi_2 \hat{f}_{2h}(x) \quad ?$$

$$\Rightarrow \pi_1 \frac{1}{n_1} \sum_{i=1}^{n_1} \frac{1}{h^d} K\left(\frac{x - X_{1i}}{h}\right) > \pi_2 \frac{1}{n_2} \sum_{i=1}^{n_2} \frac{1}{h^d} K\left(\frac{x - X_{2i}}{h}\right)$$

1.2 Behavior for very small h : Reduction to 1-NN

Suppose that X_{11} is the observation nearest to x , i.e.

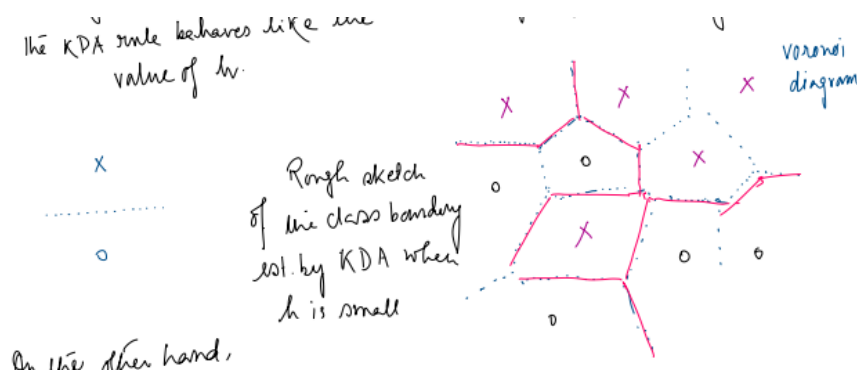
$$\|x - X_{11}\| < \|x - X_{ij}\| \quad \text{for any other choice of } i = 1, 2, j = 1, 2, \dots, n_i.$$

Then

$$\frac{K\left(\frac{x - X_{ij}}{h}\right)}{K\left(\frac{x - X_{11}}{h}\right)} = \frac{C e^{-\frac{1}{2h^2}\|x - X_{ij}\|^2}}{C e^{-\frac{1}{2h^2}\|x - X_{11}\|^2}} = e^{-\frac{1}{2h^2}\{\|x - X_{ij}\|^2 - \|x - X_{11}\|^2\}} \rightarrow 0 \quad \text{as } h \rightarrow 0.$$

So, for any fixed sample size, the KDA rule behaves like the **1-NN classifier** if we use a very small value of h .

Rough sketch of the class boundary estimated by KDA with small h (left), compared with the corresponding Voronoi diagram (right).



1.3 Behavior for large bandwidth h

On the other hand, if the priors are unequal ($\pi_1 \neq \pi_2$), for a large bandwidth h , the KDA rule will classify *all* observations to the class having the larger prior.

What if the priors are equal?

$$\text{Bias } \hat{f}_h(x) = O(h^2)$$

1.4 Mean and Variance of the KDE

$$\hat{f}_h(x) = \frac{1}{nh^d} \sum_{i=1}^n e^{-\frac{1}{2} \frac{\|x - X_i\|^2}{h^2}} \quad (\text{KDE})$$

$$\text{Var } \hat{f}_h(x) = O\left(\frac{1}{nh^d}\right) \quad (\text{under the assumption that } h \text{ is small})$$

$$E\{\hat{f}_h(x)\} = E\left[\frac{1}{h^d} e^{-\frac{1}{2} \frac{\|x - X_1\|^2}{h^2}}\right]$$

$$= \int \frac{1}{h^d} \left(e^{-\frac{1}{2} \frac{\|x - x_1\|^2}{h^2}} \right) f(x_1) dx_1 \quad (\text{true underlying density func. } X_1 \sim f)$$

$$= \frac{1}{h^d} \int \left(1 - \frac{1}{2} \frac{\|x - x_1\|^2}{h^2} + O(h^{-4}) \right) f(x_1) dx_1$$

$$= \frac{1}{h^d} \left[1 - \frac{1}{2h^2} E\|x - X_1\|^2 + O(h^{-4}) \right] \quad (\text{assume existence of moments})$$

$$\text{Var}[\hat{f}_h(x)] = \dots \quad \left[\text{We know that } \text{Var } \hat{f}_h(x) \downarrow 0 \text{ as } h \uparrow \infty \right]$$

1.5 The KDA Rule and the Sign Function $\beta_h(x)$

$$\delta_{KDA}(x) = \begin{cases} 1 & \text{if } \pi_1 \hat{f}_{1h}(x) > \pi_2 \hat{f}_{2h}(x) \\ 2 & \text{o.w.} \end{cases}$$

Now, $\pi_1 \hat{f}_{1h}(x) > \pi_2 \hat{f}_{2h}(x)$

$$\beta_h(x) = \pi_1 \hat{f}_{1h}(x) - \pi_2 \hat{f}_{2h}(x) > 0.$$

$$\text{If } h \uparrow \infty, \quad \text{Var}(\beta_h(x)) \longrightarrow 0, \quad |\beta_h(x) - E\beta_h(x)| \longrightarrow 0.$$

1.6 Asymptotic sign of $\beta_h(x)$ for large h

So, for large h , the KDA rule is determined by the sign of $E(\beta_h(x))$:

$$\begin{aligned} & E \left[\pi_1 \frac{1}{h^d} K \left(\frac{x - X_{11}}{h} \right) - \pi_2 \frac{1}{h^d} K \left(\frac{x - X_{21}}{h} \right) \right] \\ & \sim E \left[\pi_1 K \left(\frac{x - X_{11}}{h} \right) - \pi_2 K \left(\frac{x - X_{21}}{h} \right) \right] \\ & \quad \pi_1 \left(1 - \frac{1}{2h^2} E\|x - X_{11}\|^2 + O(h^{-4}) \right) \\ & \quad - \pi_2 \left(1 - \frac{1}{2h^2} E\|x - X_{21}\|^2 + O(h^{-4}) \right) \end{aligned}$$

$$\approx (\pi_1 - \pi_2) - \frac{1}{2h^2} \left\{ \pi_1 E\|x - X_{11}\|^2 - \pi_2 E\|x - X_{21}\|^2 \right\} + O(h^{-4})$$

- If $\pi_1 \neq \pi_2$, it is **positive** if $\pi_1 > \pi_2$ and **negative** if $\pi_1 < \pi_2$.
- If $\pi_1 = \pi_2$, the sign depends on whether $E\|x - X_{11}\|^2 \gtrless E\|x - X_{21}\|^2$.
 - If $E\|x - X_{11}\|^2 > E\|x - X_{21}\|^2$, it turns out to be **negative**.
 - If $E\|x - X_{11}\|^2 < E\|x - X_{21}\|^2$, it turns out to be **positive**.

where $X_{11} \sim f_1$ and $X_{21} \sim f_2$, with μ : mean vector and dispersion matrix defined below.

1.7 Explicit computation via mean vector and dispersion matrix

Suppose that μ_i and Σ_i are the mean vector and dispersion matrix of the i -th class ($i = 1, 2$).

$$E\|x - X_{11}\|^2 = \|x - \mu_1\|^2 + \text{trace}(\Sigma_1)$$

Derivation:

$$\begin{aligned} E\|x - X_{11}\|^2 &= E(x - X_{11})'(x - X_{11}) = E \text{tr}(x - X_{11})'(x - X_{11}) \\ &= E \text{tr}(X_{11} - x)(X_{11} - x)' \quad \underbrace{X_{11}}_{\text{rv}} \quad \underbrace{x}_{\text{fix}} \\ &= \text{tr} E(X_{11} - x)(X_{11} - x)' \end{aligned}$$

$$= \text{tr} \left[E(X_{11} - \mu_1)(X_{11} - \mu_1)' + E(x - \mu_1)(x - \mu_1)' - 2E(X_{11} - \mu_1)(x - \mu_1)' \right]$$

Side computation:

$$\begin{aligned} X_{11} - x &= (X_{11} - \mu_1) - (x - \mu_1) \\ (X_{11} - x)(X_{11} - x)' &= (X_{11} - \mu_1)(X_{11} - \mu_1)' + (x - \mu_1)(x - \mu_1)' - 2(X_{11} - \mu_1)(x - \mu_1)' \\ &= \text{tr}\{\Sigma_1 + \|x - \mu_1\|^2\} = \|x - \mu_1\|^2 + \text{tr}(\Sigma_1) \end{aligned}$$

Hence:

$$E\|x - X_{11}\|^2 = \|x - \mu_1\|^2 + \text{tr}(\Sigma_1), \quad E\|x - X_{21}\|^2 = \|x - \mu_2\|^2 + \text{tr}(\Sigma_2)$$

So the sign of $\beta_h(x)$ turns out to be the same as that of

$$\left\{ \|x - \mu_2\|^2 + \text{tr}(\Sigma_2) - \|x - \mu_1\|^2 - \text{tr}(\Sigma_1) \right\}$$

1.8 KDA reduces to a linear classifier

So, for the KDA classifier,

$$\left\{ \|x - \mu_2\|^2 - \|x - \mu_1\|^2 \right\} + \left\{ \text{tr}(\Sigma_2) - \text{tr}(\Sigma_1) \right\} \text{ is positive} \implies \text{classify to class 1.}$$

Clearly, this is a **linear classifier**. If $\Sigma_1 = \Sigma_2$, it will classify an observation x to class 1 if

$$\|x - \mu_2\|^2 > \|x - \mu_1\|^2.$$

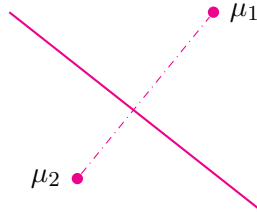


Figure 1: Linear separating boundary between μ_1 and μ_2 when $\Sigma_1 = \Sigma_2$.

2 Nearest Neighbor Density Estimation

2.1 Classification using nearest neighbor density estimates

$$\pi_1 \hat{f}_1(x) \geq \pi_2 \hat{f}_2(x)$$

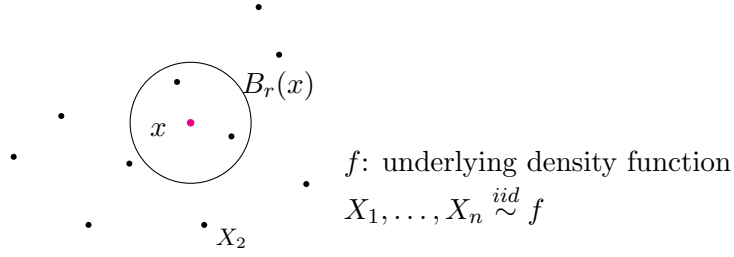


Figure 2: Scatter of i.i.d. observations with ball $B_r(x)$ used for nearest-neighbor density estimation.

2.2 Motivating the estimator

$$\forall \varepsilon > 0 \exists \delta \text{ s.t. if } \|x - x^*\| < \delta, |f(x) - f(x^*)| < \varepsilon.$$

If f is assumed to be nearly constant on $B_r(x)$, then

$$P(B_r(x)) = \int_{y \in B_r(x)} f(y) dy \approx f(x) \int_{y \in B_r(x)} 1 dy = f(x) \cdot \text{Vol}(B_r(x))$$

If K out of n observations lie in $B_r(x)$, then K/n can serve as an estimate of $P(B_r(x))$.

This leads to an estimate of the density function f , given by

$$\hat{f}(x) = \frac{K/n}{\text{Vol}(B_r(x))}$$

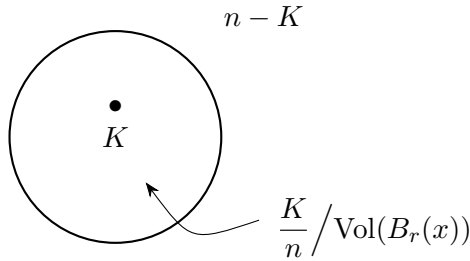


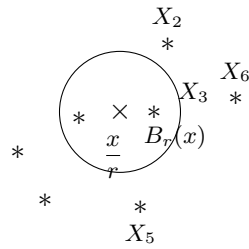
Figure 3: K observations out of n inside the ball $B_r(x)$.

2.3 Loftsgaarden & Quesenberry (1955) estimator

Nearest neighbor density estimation (Loftsgaarden & Quesenberry, 1955).

If K out of n observations are made in $B_r(x)$, then we can use

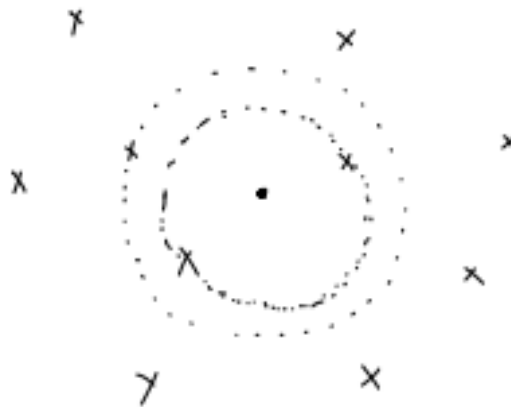
$$\hat{f}(x) = \frac{K/n}{\text{Vol}(B_r(x))} \quad : r \text{ is fixed; } K \text{ is random.}$$

Figure 4: Points $X_1, \dots, X_6$ and ball $B_r(x)$.

Loftsgaarden & Quesenberry suggested considering the **smallest neighborhood (ball)** that contains K neighbors of x .

$$\hat{f}_K(x) = \frac{K/n}{\text{Vol}(B_{r_K}(x))}, \quad \text{where } r_K \text{ is the distance between } x \text{ and its } K\text{-th nearest neighbor.}$$

Smallest ball containing K nearest neighbors of x .



2.4 Choice of K

How to choose the value of K ?

Result: If f is continuous, then if $K \rightarrow \infty$ and $\frac{K}{n} \rightarrow 0$ as $n \rightarrow \infty$,

$$\hat{f}_K(x) \xrightarrow{P} f(x).$$

In practice, we can select the value of K by using the likelihood cross-validation (CV) method, where we maximize the **pseudo-likelihood**

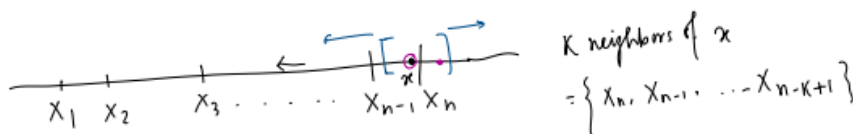
$$\prod_{i=1}^n \hat{f}_{K,-i}(X_i)$$

where $\hat{f}_{K,-i}$ is the density estimate constructed based on $(n-1)$ observations, leaving X_i out of the training data ($i = 1, \dots, n$).

$$\hat{f}_K(x) = \frac{K/n}{\text{Vol}(B_{r_K}(x))}, \quad \text{where } r_K \text{ is the distance between } x \text{ and its } K\text{-th nearest neighbor.}$$

2.5 A boundary issue – one-dimensional illustration

One-dimensional illustration: K neighbors of x .



$$\begin{aligned}\hat{f}_K(x) &= \frac{K/n}{2|x - X_{n-K+1}|}, \quad x \geq X_n \\ \int_{-\infty}^{+\infty} \hat{f}_K(x) dx &\geq \int_{X_n}^{\infty} \hat{f}_K(x) dx = \frac{K/n}{2} \int_{X_n}^{\infty} \frac{1}{|x - X_{n-K+1}|} dx \\ &= \frac{K/n}{2} \int_a^{\infty} \frac{1}{y} dy \quad \text{for some } a > 0. \\ &= \infty.\end{aligned}$$

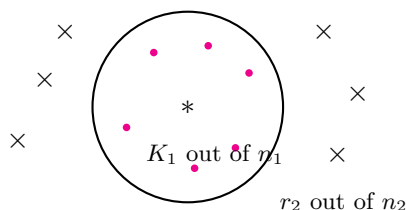


Figure 5: Illustration of the ball containing K_1 class-1 and K_2 class-2 observations.

$$\pi_1 f_1(x) \geq \pi_2 f_2(x) \implies \frac{n_1}{n} \cdot \frac{K_1/n_1}{\text{Vol}(\text{Ball})} \geq \frac{n_2}{n} \cdot \frac{K_2/n_2}{\text{Vol}(\text{Ball})} \iff K_1 \geq K_2$$

We can use the distance between x and its K -th neighbor in the sample as the radius of the ball.

3 K-Nearest Neighbor Classification

In that case, the classifier looks at the K nearest neighbors of x and assigns the observation to the class which has the maximum number of representatives in the set of K neighbors.

This classifier is known as the **K -Nearest Neighbor Classifier**.

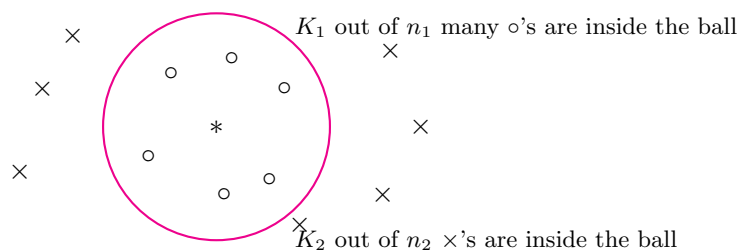


Figure 6: K -NN classification: counting representatives of each class within the ball.

$$\begin{aligned}
 P(\text{class 1} \mid x) &= \frac{\pi_1 f_1(x)}{\pi_1 f_1(x) + \pi_2 f_2(x)} \\
 &\approx \frac{\frac{n_1}{n} \cdot \frac{K_1}{n_1} \cdot \frac{1}{\text{Vol}(\text{Ball})}}{\frac{n_1}{n} \cdot \frac{K_1}{n_1} \cdot \frac{1}{\text{Vol}(\text{Ball})} + \frac{n_2}{n} \cdot \frac{K_2}{n_2} \cdot \frac{1}{\text{Vol}(\text{Ball})}} = \frac{K_1}{K_1 + K_2}
 \end{aligned}$$

So, this classifier basically classifies an observation to the class for which the estimated posterior probability is maximum.

For K -nearest neighbor classification, K is usually chosen by **cross-validation**.

Result: If the underlying class densities are continuous, and

- (i) $K \rightarrow \infty$, and
- (ii) $\frac{K}{n} \rightarrow 0$ as $n \rightarrow \infty$,

then the error rate/misclassification probability of the K -nearest neighbor classifier converges in probability to the Bayes risk (i.e., the misclassification probability of the Bayes classifier).

4 High-Dimensional Behavior of the 1-NN Classifier

4.1 Setup

Now, consider a binary classification problem.

$$\begin{aligned}
 \text{Class-1: } & N_d(0, \sigma_1^2 I), \quad X_{11}, \dots, X_{1n_1} \\
 \text{Class-2: } & N_d\left(\begin{pmatrix} \mu \\ \mu \\ \vdots \\ \mu \end{pmatrix}, \sigma_2^2 I\right), \quad X_{21}, \dots, X_{2n_2}
 \end{aligned}$$



How will the 1-NN classifier work in this case when d is large?

4.2 Scaled squared distances

$$\frac{\|X_1 - X_2\|^2}{d} = \frac{(X_1^{(1)} - X_2^{(1)})^2 + (X_1^{(2)} - X_2^{(2)})^2 + \dots + (X_1^{(d)} - X_2^{(d)})^2}{d} \longrightarrow 2\sigma_1^2 \quad \text{as } d \rightarrow \infty$$

where

$$X_1, X_2 \sim N_d(0, \sigma_1^2 I), \quad Y_1, Y_2 \sim N_d(\mu \mathbf{1}, \sigma_2^2 I).$$

$$X_1^{(1)} \sim N(0, \sigma_1^2), \quad X_2^{(1)} \sim N(0, \sigma_1^2), \quad X_1^{(1)} - X_2^{(1)} \sim N(0, 2\sigma_1^2)$$

$$\frac{(X_1^{(1)} - X_2^{(1)})^2}{2\sigma_1^2} \sim \chi_1^2$$

$$\frac{\|Y_1 - Y_2\|^2}{d} \longrightarrow 2\sigma_2^2$$

$$\frac{\|X_1 - Y_1\|^2}{d} \longrightarrow \sigma_1^2 + \sigma_2^2 + \mu^2 \quad (\text{check})$$

Scaled sq. distribution	$Z \sim N(0, \sigma_1^2 I)$	$Z \sim N(\mu, \sigma_2^2 I)$
Sq. dist from obs. in Class 1	$2\sigma_1^2$	$\sigma_1^2 + \sigma_2^2 + \mu^2$
Sq. dist from obs. in Class 2	$\sigma_1^2 + \sigma_2^2 + \mu^2$	$2\sigma_2^2$

Table 1: Limiting scaled squared distances in the high-dimensional regime.

$$\|Z - X_Y\|^2/d < \|Z - X_{21}\|^2/d$$

4.3 Condition for correct classification

For correct classification of an observation from Class 1, we need:

$$2\sigma_1^2 < \sigma_1^2 + \sigma_2^2 + \mu^2 \implies \sigma_1^2 - \sigma_2^2 < \mu^2$$

For correct classification of an observation from Class 2, we need:

$$\sigma_2^2 - \sigma_1^2 < \mu^2$$

So, eventually we need

$$\tilde{\mu}^2 > |\sigma_1^2 - \sigma_2^2|$$

4.4 General limiting quantities

For general parameters (μ_1, Σ_1) and (μ_2, Σ_2) , define:

$$\tilde{\mu}^2 = \lim_{d \rightarrow \infty} \frac{\|\mu_1 - \mu_2\|^2}{d}, \quad \tilde{\sigma}_1^2 = \lim_{d \rightarrow \infty} \frac{\text{tr}(\Sigma_1)}{d}, \quad \tilde{\sigma}_2^2 = \lim_{d \rightarrow \infty} \frac{\text{tr}(\Sigma_2)}{d}$$